

Sreeniketh Aathreya

Software Engineer / Solutions Architect

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Professional Summary

- Outcome-driven Solutions Architect with over 7 years of comprehensive experience in application and ecosystem architecture, AI-native solution design, Python automation, DevOps practices, ML/AI, Internet Technologies, Data Science, and Cloud technologies with proven leadership capabilities in enterprise environments.
- Extensive experience in designing and implementing end-to-end scalable applications, automation frameworks and cloud-based solutions across multiple industries, including real estate, fintech, healthcare and e-Commerce.
- Expert knowledge and proficiency in modern web technologies including React, HTML5, CSS3, Node.js, and RESTful API development and GraphQL integrations.
- Worked extensively with Object-Oriented Programming concepts in Python, Java, Kotlin, and GOLANG, implementing robust software architecture patterns and design principles in enterprise applications.
- Deep expertise in the cloud computing ecosystem: AWS (EC2, RDS, Lambda, CloudFormation, ELB), GCP (Master Tier in Vertex AI, Gemini Enterprise, BigQuery), container and Kubernetes orchestration, and Infrastructure as Code with Terraform.
- Comprehensive experience in Data Engineering with structured and unstructured data processing, ETL pipeline development, data validation, and advanced analytics using Python and distributed computing frameworks.
- Hands-on experience with cutting-edge AI/ML libraries and frameworks including Anthropic, Gemini Enterprise, Google ADK, LangChain, TensorFlow, and implementing AI-native systems for solving enterprise problems with advanced ML techniques including natural language processing, computer vision, reinforcement learning, and implementing socially-aware AI systems with 98+ % accuracy rates. Appointed as AI Labs Lead for the expertise shown in implementing lean, creative AI solutions and the exhibiting rigor in scaling them out.
- Proficient in distributed systems, blockchain technology with R3 Corda platform, and NoSQL databases with expertise in applying data optimization techniques in B2B and B2C enterprise solutions.
- Strong understanding of CI/CD pipelines, DevOps methodologies, and have successfully implemented automated deployment strategies using GitHub Actions, Jenkins, and Kubernetes across multiple cloud platforms.
- Extensive experience with security frameworks, vulnerability assessment, DDoS mitigation, and implementing enterprise-grade access control systems for sensitive data governance.
- Deep knowledge of SDLC, Agile, and Scrum methodologies with proven track record of leading cross-functional teams and delivering multi-million dollar projects well within allocated time and budget.
- Strong background in database design and optimization across multiple platforms including PostgreSQL, MySQL, data lakes and vector databases.
- Solid track record in leading solution discovery workshops and translating client business goals into technical architectures, staffing estimates, and delivery-ready proposals, with sharp sensitivity to business pain points and proven ability to drive outcomes as an individual contributor and as a skilled leader.
- Experienced in build-versus-buy evaluation and recommending best-fit cloud and AI toolsets across GCP, AWS, and Azure to align technical solutions with client objectives and delivery constraints.
- Proficient in automation testing frameworks, security audit scripting, and system log analysis that reduced vulnerabilities by 30% in enterprise environments.
- Strong background in technical sales support and customer advocacy, with proven track record of closing \$10+ million enterprise deals through comprehensive technical demonstrations, proof-of-concepts (POCs), and customer analysis reports (CARs), and setting up trade-off comparisons and visualizations for value propositions that aligned technical solutions with business objectives.
- Proven expertise in presenting technical topics to diverse audiences, including C-level executives, IT leadership, enterprise business professionals, and technical stakeholders, with experience in leading sales activities and facilitating product adoption for complex software solutions.
- Deep experience in solution architecture and technical proposal development, with comprehensive understanding of global platform technologies and edge computing solutions that make internet and application experiences faster, more reliable, and more secure for compliance-sensitive organizations.
- **F1/OPT, H1B lottery picked**

Technical Skills

Languages and Frameworks: Python, Go(GOLANG), JavaScript, Bash, React, Kotlin, SQL, C, C++, Java, TypeScript, YAML

Operating Systems: Linux, Windows, Unix, Ubuntu.

Web Technologies: JavaScript, HTML, RShiny, XML, CSS, PHP, WordPress, Spring, RabbitMQ, React, Node.js, FastAPI

Database Management: MS Access, MySQL, Oracle, PostgreSQL, RESTful APIs, GraphQL

Cloud Computing Tools: AWS CLI, EC2, RDS, ELB, Elastic Cache, CloudFormation, CloudWatch, GCP CLI, Cloud Run, App Engine, Secret Manager, AI Model Garden, Vertex AI, BigQuery, Prometheus, Grafana, Git-Lab, Docker, Ansible Tower, Terraform, Jenkins, Helm, GitHub, Databricks, Docker, Github Actions, NodeJS, Kubernetes, Aurora.

AI Integrations: Google ADK, Anthropic, Llama3, Gemini Enterprise, LangChain

Architecture: Microservices, API-first design, event-driven systems, solution diagramming, build-vs-buy evaluation

Other: TCP/IP, DNS, TLS/SSL, HTTP/HTTPS, CDN Architecture, Edge Computing, Global Platform Technologies, Intelligent Edge, Bot Defender, WAF. HIPAA Compliance, FHIR Standards, Healthcare Interoperability, Medical Device Security, Regulatory Compliance, Technical Solution Design, System Integration, Performance Optimization, Scalability Planning, Enterprise Architecture.

Certifications

GCP Certified Professional Data Engineer : Credly(2026-01-16 to 2028-01-16)

GCP Certified Professional Machine Learning Engineer : Credly(2025-11-26 to 2027-11-26)

AWS Certified AI Practitioner : Credly and Certmetrics(2024-09-30 to 2027-09-30)

AWS Certified Developer – Associate : Credly and AWS(2021-09-03 to 2024-09-03)

AWS Certified Solutions Architect – Associate : Certmetrics(2020-12-12 to 2023-12-12)

AWS Certified Data Engineer – Associate

Corda Certified Developer

Databricks Lakehouse Fundamentals : Databricks(2024-08-27 to 2026-08-27)

Databricks Generative AI Fundamentals Databricks(2024-08-26 to 2026-08-26)

Work Experience

Solution Architect & AI Labs Lead HatchWorks AI – Remote, US

November 2025 - Present

- Partnered with the CTO to shape the technical direction of 20+ client engagements from discovery through delivery, translating business goals into delivery-ready technical architectures, staffing estimates, and scoped proposals for AI-native and cloud-native platforms.
- Supported 30+ pre-delivery engagements, collaborating with multiple enterprise strategy, product and client sales partner teams to chart and accelerate roadmaps and pre-sales engineering processes.
- Designed AI-native workflows leveraging foundation models, embeddings, agentic orchestration, and LLM integration patterns (LangChain, Vertex AI) to deliver intelligent, purpose-built software products for enterprise clients across multiple industries.
- Participated in multiple early-stage prototyping and architecture handoffs during project kickoff, ensuring delivery teams have a clear and stable technical foundation spanning full-stack development, data engineering, and agentic engineering.
- Evaluated various novel implementations to recommend best-fit cloud and AI toolsets for client engagements, contributing to reusable architectural patterns, delivery accelerators, and reference implementations that improved organizational delivery velocity.
- Major contributor in establishing **GenDD(Click here)** - an organization-wide methodology and AI delivery automation framework for performing 150+ tasks that fall under the umbrella of software development, the adoption of which accelerated delivery timelines. 347 AI Hrs Reclaimed / Month. 75% productivity gain while maintaining <3% defect rate.
- Environment: Python, TypeScript, React, Node.js, FastAPI, LangChain, GCP (Vertex AI, BigQuery), AWS, Docker, Terraform, CI/CD, Kubernetes, Gemini, Anthropic, Agile/Scrum.

Software Engineer, i-Link Solutions – Chantilly, VA

Dec 2024 - November 2025

- Led technical solutions development for Lantern, a critical healthcare monitoring platform serving enterprise healthcare customers nationwide, working directly with healthcare providers, payers, and government health agencies to deliver fast, reliable, and secure healthcare technology solutions.
- Conducted technical demonstrations and solution presentations to diverse healthcare stakeholders, including C-level executives, IT leadership, and clinical teams, successfully advocating for technical solutions that align healthcare interoperability requirements with business objectives.
- Architected comprehensive event-driven, cloud-native technical solutions that process millions of healthcare API calls daily with 99.9% uptime, implementing advanced security protocols, performance optimization, and compliance frameworks including HIPAA and 21st Century Cures Act requirements.
- Managed technical proposal processes and customer analysis reports for healthcare enterprise accounts, ensuring accurate technical recommendations that address customer requirements while maximizing product value through scalable platform solutions.
- Collaborated with cross-functional teams including product engineering, cloud architects, and government stakeholders to design technical solutions that meet complex healthcare regulatory requirements and enterprise performance standards.

- Mentored junior engineers on architecture patterns, microservices design, and Agile delivery practices, contributing to improved team velocity and engineering maturity across the organization.
- Tracked account activity and business analysis for healthcare enterprise customers, providing technical insights and solution recommendations that support account growth and long-term customer success.
- Environment: Go, Python, PostgreSQL, RabbitMQ, Docker, Kubernetes, Microservices, Automated Monitoring, Healthcare IT, Open Source Development. **(Lantern)**

Pre-Sales Engineer, R3 – Manhattan, NY

June 2023 - Apr 2024

- Led pre-sales technical activities and customer-facing solution demonstrations that resulted in closing a \$4 million enterprise deal with major financial services clients, conducting comprehensive technical presentations to C-level executives and IT leadership teams.
- Designed and executed proof-of-concepts (POCs) for enterprise blockchain solutions, creating detailed customer analysis reports and technical proposals that aligned cutting-edge technology capabilities with specific business requirements and regulatory compliance needs.
- Managed technical demonstration processes for distributed ledger platforms, presenting complex technical concepts to diverse audiences including technical teams, business stakeholders, and compliance officers across multiple enterprise customers.
- Evaluated and recommended cloud and AI toolset strategies (AWS vs. GCP vs. Azure) for client engagements, performing build-versus-buy assessments that optimized delivery timelines and reduced infrastructure costs.
- Collaborated with sales teams to identify technical opportunities and drive account growth, providing technical expertise and solution architecture that supported enterprise sales cycles and customer adoption initiatives.
- Developed reusable architectural patterns, reference implementations, and comprehensive technical documentation for enterprise customers, ensuring accurate implementation guidance for complex multi-cloud deployments.
- Environment: Kotlin, Python, Go, R3 Corda, AWS, GCP, Kubernetes, Docker, Terraform, Jenkins, GitHub Actions, Distributed Systems, Infrastructure as Code

Solutions Architect, Akamai Technologies Inc. – Bengaluru, India

Nov 2021 - July 2022

- Drove pre-sales strategy initiatives by designing and demonstrating cutting-edge services that address complex web performance and security challenges for global enterprise customers, utilizing unified CDN and security platform capabilities to secure and influence \$20M+ in enterprise deals.
- Led technical demonstrations and proof-of-concepts for global enterprise customers, presenting Akamai's technology solutions including CDN optimization, DDoS mitigation, Web Application Firewall (WAF), and performance enhancement to diverse technical and business audiences.
- Created comprehensive customer analysis reports and technical proposals for enterprise accounts, providing detailed build-versus-buy recommendations that align Akamai's global platform capabilities with customer business objectives and technical requirements.
- Managed technical proposal processes for large-scale enterprise deployments, ensuring accurate technical information and solution architecture that addressed customer performance, security, and reliability requirements.
- Collaborated with product engineering and customer success teams to deliver tailored solutions that met specific customer requirements while maintaining platform scalability and security standards for global enterprise deployments.
- Environment: CDN Technologies, DDoS Mitigation, Web Performance Optimization, Security Analysis, Automation Scripting, Global Infrastructure Management.

Application Security Engineer, Cognizant Technology Solutions – Bengaluru, India

Aug 2020 – Nov 2021

- Provided technical consulting and security solution recommendations to enterprise customers, conducting comprehensive security assessments and vulnerability analysis that reduced system vulnerabilities by 30% across multiple enterprise environments.
- Developed customer-facing technical presentations and security solution demonstrations, presenting complex cybersecurity concepts to diverse audiences including IT leadership, security teams, and business stakeholders.
- Collaborated with customer technical teams to implement security best practices and compliance frameworks, providing technical guidance and solution architecture that aligned security requirements with business objectives.
- Created technical documentation and security assessment reports for enterprise customers, ensuring accurate technical recommendations and implementation guidance for security enhancement initiatives.
- Environment: Linux Administration, Python, DevOps, CI/CD Pipelines, Security Auditing, Vulnerability Assessment, Penetration Testing, Automation Scripting, Incident Response.

Education

- **Rutgers University**, MS in Computer Science **GPA: 3.875/4.0**
- **PES University**, B.Tech in Computer Science **GPA: 3.5/4.0**

Sept 2022 - May 2024

Aug 2016 - June 2020

Paid Side-Projects

AI Security Researcher, Ackuity.ai – Fairfax, VA

Sep 2024 - Jan 2025

- Spearheaded the development of enterprise-grade access control frameworks for Amazon Bedrock, implementing sophisticated metadata filtering and role-based authentication systems that enable secure doctor-patient data governance while maintaining full LLM functionality and reducing unauthorized data access risks by 90%.
- Developed enterprise-grade security platform using full-stack architecture, implementing React-based dashboards and Python backend services for Amazon Bedrock integration that increased developer productivity by 200%.
- Technical papers published on ([click here - Medium](#)) and ([click here - LinkedIn](#)) with a combined total of 3000 reads.
- Conducted extensive vulnerability assessments and penetration testing on LLM systems, identifying and mitigating potential security threats in AI-powered healthcare applications while ensuring HIPAA compliance and data privacy regulations.
- Environment: Amazon Bedrock, Python, AI Security, LLM Security, Healthcare Data Governance, HIPAA Compliance, Vulnerability Assessment, Technical Writing.

Lead Software Developer, Amore Nidra Inc. – Los Angeles, CA
([Click here](#))

Sept 2024 - Dec 2024

- Architected and developed complete e-commerce platform from scratch, implementing comprehensive monitoring and alerting systems that reduced page load times by 60% and created streamlined automated workflows resulting in 70% faster operational processes.
- Designed and implemented custom Shopify applications and seamless third-party integrations using Node.js, GraphQL APIs, and advanced JavaScript frameworks, delivering measurable business impact with 45% increase in average order value and remarkable 200% increase in online sales revenue.
- Implemented automated performance monitoring and real-time analytics dashboards that enhanced system reliability and user engagement, demonstrating end-to-end system lifecycle management.
- Built comprehensive testing framework with unit, integration, and end-to-end tests, maintaining 95%+ code coverage and automated CI/CD deployment through GitHub Actions.
- Environment: Shopify, Liquid, Node.js, GraphQL, JavaScript, CSS3, Payment Gateway APIs, E-commerce Analytics, SEO Optimization, Customer Analytics.

Projects

[Click here for Github](#)

- **AI-Native Medical Content Platform:** Architected and delivered full-stack AI-native application using Python/FastAPI backend with React/TypeScript frontend, integrating Anthropic foundation models for automated content generation. Designed cloud-native architecture with comprehensive technical documentation and CI/CD deployment.([Click here](#))
- **RAG-Powered E-commerce Solution:** Built scalable, event-driven e-commerce platform leveraging LLM orchestration with embeddings-based retrieval for conversational product search across 15,000+ items. Implemented agentic workflow architecture with Python backend, TypeScript frontend, and integrated foundation models for intelligent user interactions.([Click here](#))
- **Enterprise Agentic AI Framework:** Designed comprehensive solution architecture for multi-agent AI system using Model Context Protocol, featuring stateful LLM orchestration with multi-agent coordination patterns, RESTful API design, and enterprise-grade security. Created detailed technical documentation and implementation guides for client delivery.([Click here](#))
- **Cloud-Native AI Security Platform:** Architected distributed, scalable system for AI governance with role-based access control, implementing Infrastructure-as-Code with Terraform. Delivered complete solution including technical specifications, architecture diagrams, and deployment automation for enterprise clients.([Click here](#))